Quiz 2 — Sample 1

DATA 412/612

Weeks 1–9. The new material is tidyr, mutating and filtering joins, databases, and strings.

Write your answers in an R script. You may use R and RStudio. You may use your notes and the course website (`https://amousavi1.github.io/612/data-612.html`). You may not use Google, any other website, or an LLM. Load the tidyverse, `nycflights13`, `DBI`, `RSQLite`, and `dbplyr`.

1. Add the airline name from `airlines` onto `flights`. How many flights still have a missing airline name?

2. How many flights have a `tailnum` that does not appear in `planes`? Use a filtering join, not a `filter()` on a hand-built list of tail numbers.

3. Use an in-memory SQLite database, as in the databases lecture. Copy the tibble below into the database, then from that remote table keep rows with `score` of at least 8, show the SQL, collect the result into a local tibble, and disconnect.

```
labs <- tibble(
  name = c("Ada", "Lin", "Grace", "Alan"),
  score = c(9, 7, 8, 6)
)
```

4. On the character vector below, use one regular expression to replace every token that starts with `AA` or `DL`, followed by a dash and three digits, with the word `hit`. Leave the other strings unchanged.

```
c("AA-101", "UA-22", "DL-303", "B6-1", "aa-404")
```